INDUSTRIAL TRAINING REPORT

ON

An Edge-Intelligent System for Continuous Ventricular Arrhythmia Detection

Edge-AI Pipeline for Real-Time Cardiac Monitoring

At Centre of Excellence in Affordable Healthcare (CoEAH), IIT Kharagpur

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

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Certificate

This is to certify that **Mr. Mahir Desai** (Registration No. 229302198) has successfully completed the Industrial Training during the period from **May 2026** to **July 2026** at the **Centre of Excellence in Affordable Healthcare (CoEAH), Indian Institute of Technology (IIT) Kharagpur**, in partial fulfillment of the requirements for the degree of **Bachelor of Technology in Computer Science and Engineering** from **Manipal University Jaipur**, during the academic year 2025–2026.

He worked on the project titled *"An Edge-Intelligent System for Continuous Ventricular Arrhythmia Detection"* and was trained in the domains of Edge Artificial Intelligence, Digital Signal Processing, Embedded Systems (ESP32/FreeRTOS), and Real-Time Biomedical Firmware Development.

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Place: Kharagpur

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Dr. Debanjan Das
Mentor & Project Supervisor
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IIT Kharagpur

Note: The original certificate issued by IIT Kharagpur / KRITI Initiative is attached as an appendix.

Declaration

I hereby declare that the Industrial Training Report entitled **"An Edge-Intelligent System for Continuous Ventricular Arrhythmia Detection"** submitted to the **Department of Computer Science and Engineering, Manipal University Jaipur**, is an authentic record of my own work carried out during the period from **May 2026** to **July 2026** at the **Centre of Excellence in Affordable Healthcare (CoEAH), IIT Kharagpur**, under the external mentorship of **Dr. Debanjan Das** and co-guidance of **Sayantani Banerjee** (PhD Scholar, IIT Kharagpur).

The matter presented in this report has not been submitted elsewhere for the award of any other degree or diploma.

Date: _____

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Certified that the above statement made by the student is correct to the best of our knowledge and belief.

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Head of Department
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I also extend my heartfelt thanks to **Sayantani Banerjee** (PhD Scholar, IIT Kharagpur), whose expertise in Edge-AI, biomedical signal processing, and firmware architecture proved instrumental in refining the system. I sincerely thank the **KRITI Initiative** for providing this extraordinary research pathway at IIT Kharagpur.

I would like to express my sincere thanks to **Prof. Dr. Neha Chaudhary** (Head, Department of Computer Science and Engineering, Manipal University Jaipur), **Dr. Susheela Vishnoi** (Course Coordinator, Industrial Training), and the faculty of MUJ whose support facilitated the smooth execution of this training.

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Abstract

Malignant ventricular arrhythmias—Ventricular Tachycardia (VT) and Ventricular Fibrillation (VF)—are the primary electrophysiological triggers for Sudden Cardiac Death (SCD). Continuous ambulatory monitoring is critical, yet existing wearable devices suffer from cloud-dependency latency, overwhelming false alarms from motion artifacts, and diagnostic degradation from aggressive model quantization.

This report presents the design and implementation of an end-to-end edge-intelligent wearable system. The framework integrates dual-channel analog bio-potential acquisition (AD8232 + ESP32-S3) at 250 Hz, a Signal Quality Assessment (SQA) gate with 15-second inference lockout, zero-phase 4th-order Butterworth filtering, and a compact 1D Convolutional Neural Network (1D-CNN) deployed in full floating-point (FP32) precision directly on-chip without lossy quantization. A native Android application (Kotlin, Jetpack Compose) provides real-time BLE 5.0 telemetry, waveform visualization, and Heart Rate Variability (HRV) analytics.

Rigorous validation on strictly unseen inter-patient splits demonstrated an overall accuracy of 98.54%, VT/VF sensitivity of 97.46%, and specificity of 99.72%, with sub-2 ms inference latency. The working prototype was presented at the PRAVAH 2026 Innovation Fair at IIT Kharagpur.

The embedded firmware was implemented in C++ using TensorFlow Lite Micro and FreeRTOS. Model training and validation used Python (TensorFlow/Keras, SciPy, Scikit-learn). The mobile application was developed in Kotlin with Android Jetpack Compose.

About the Training Organization

Centre of Excellence in Affordable Healthcare (CoEAH), IIT Kharagpur

The Centre of Excellence in Affordable Healthcare (CoEAH) at the Indian Institute of Technology Kharagpur is an interdisciplinary research facility focused on developing frugal, high-impact medical innovations. CoEAH integrates clinical science with edge-computing engineering, designing low-cost diagnostic systems, wearable physiological sensors, and embedded AI healthcare pipelines for underserved communities.

The KRITI Initiative

The KRITI Initiative is an academic research fellowship program at IIT Kharagpur that integrates student researchers into state-of-the-art laboratories. Through KRITI (specifically the GRISHMA summer track), students collaborate with faculty and PhD scholars on active research projects, gaining hands-on experience in prototyping and validation.

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Abbreviations

Abbreviation	Full Form
ADC	Analog-to-Digital Converter
BLE	Bluetooth Low Energy
BPM	Beats Per Minute
CNN	Convolutional Neural Network
DSP	Digital Signal Processing
ECG	Electrocardiogram
FP32	32-bit Floating Point
HRV	Heart Rate Variability
MAD	Median Absolute Deviation
PTQ	Post-Training Quantization
RMSSD	Root Mean Square of Successive Differences
SCD	Sudden Cardiac Death
SDNN	Standard Deviation of NN Intervals
SQA	Signal Quality Assessment
VF	Ventricular Fibrillation
VT	Ventricular Tachycardia

1. Introduction

1.1 Background and Motivation

Cardiovascular diseases (CVDs) remain the leading cause of mortality globally. Among acute cardiac conditions, Ventricular Tachycardia (VT) and Ventricular Fibrillation (VF) are the most lethal electrophysiological triggers for Sudden Cardiac Death (SCD). During VT, abnormal electrical impulses cause the ventricles to beat dangerously fast ($>$150 BPM). If untreated, VT degenerates into VF—the ventricles quiver chaotically, cardiac output drops to near zero, and death follows within minutes without immediate defibrillation.

The 12-lead Electrocardiogram (ECG) remains the gold standard for diagnosis but is confined to hospital settings. Because these arrhythmias are paroxysmal (occurring suddenly and transiently), they are frequently missed during routine clinical visits, creating an urgent need for continuous ambulatory monitoring.

1.2 Problem Statement

Existing ambulatory monitors (Holter monitors, smart patches, and smartwatches) suffer from critical limitations:

1. **Cloud Dependency and Latency:** IoT-based wearables transmit raw data to remote servers for analysis, introducing network delays that are unacceptable during acute cardiac events.
2. **Motion Artifacts and False Alarms:** Physical activities generate severe baseline wander and muscle noise. Standard algorithms frequently misclassify motion artifacts as VF, triggering overwhelming false-positive alarms.
3. **Quantization Degradation:** TinyML edge deployments use INT8 Post-Training Quantization (PTQ), which systematically degrades sensitivity at morphological decision boundaries.

1.3 Objectives

The primary objectives of this industrial training project were:

- Design an ultra-low-power ESP32-S3 edge node paired with dual AD8232 analog front-ends for continuous ECG acquisition.
- Implement a Signal Quality Assessment (SQA) engine with a 15-second inference lockout to eliminate motion-induced false alarms.

- Deploy a compact, unquantized (FP32) 1D-CNN directly on the microcontroller's SRAM.
- Engineer a native Android application in Kotlin for real-time BLE 5.0 telemetry and clinical visualization.

1.4 Organization of the Report

Chapter 2 presents the theoretical background and literature review. Chapter 3 details the methodology including data acquisition, SQA, and preprocessing. Chapter 4 describes the implementation of the firmware and mobile application. Chapter 5 presents results and comparative analysis. Chapter 6 summarizes conclusions and future scope.

2. Background and Literature Review

2.1 Electrophysiology of Ventricular Arrhythmias

The heart's electrical conduction originates at the Sinoatrial (SA) node, propagating through the AV node and Purkinje fibers. Normal conduction manifests as the standard P-QRS-T complex on the ECG trace. In VT, ectopic ventricular pacemakers produce wide, bizarre QRS complexes at rates exceeding 150 BPM. In VF, organized depolarization ceases entirely—the waveform shows irregular, high-frequency undulations devoid of identifiable QRS complexes.

2.2 Digital Signal Processing Fundamentals

2.2.1 Butterworth High-Pass Filter

Ambulatory ECG signals are contaminated by baseline wander from respiration (0.15–0.5 Hz) and electrode impedance fluctuations. A 4th-order zero-phase Butterworth high-pass filter ($f_c = 1.0$ Hz) is employed:

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega_c}{\omega}\right)^{2N}} \tag{2.1}$$

The forward-backward filtering approach eliminates phase distortion, preserving QRS morphology.

2.2.2 Median Absolute Deviation (MAD)

Electrode displacement produces sudden voltage spikes. The Median Absolute Deviation provides an outlier-resistant metric:

$$\text{MAD} = \text{median}\left(|X_i - \text{median}(X)|\right) \tag{2.2}$$

Spikes exceeding $3.5 \times \text{MAD}$ are dynamically clamped without suppressing genuine ventricular beats.

2.3 Literature Review

Deep learning, particularly 1D-CNNs, has demonstrated strong performance for automated cardiac classification. Picon *et al.* [6] proposed a CNN-LSTM achieving 93.10% accuracy, and Barbosa *et al.* [7] achieved 96.40% with a deep CNN—however, both require cloud GPUs. Han *et al.* [8] demonstrated lightweight edge inference but used intra-

patient splitting, which inflates accuracy through data leakage. Phukan *et al.* [10] introduced SQA-aware TinyML but employed aggressive INT8 quantization, which degrades sensitivity at decision boundaries [3].

Our system resolves all three limitations by deploying an unquantized FP32 CNN entirely on-chip, incorporating motion-robust SQA, and validating exclusively on unseen patients.

3. Methodology

3.1 System Architecture

The framework connects physical analog front-ends to local edge intelligence and wireless mobile bio-feedback, as shown in Figure 3.1.

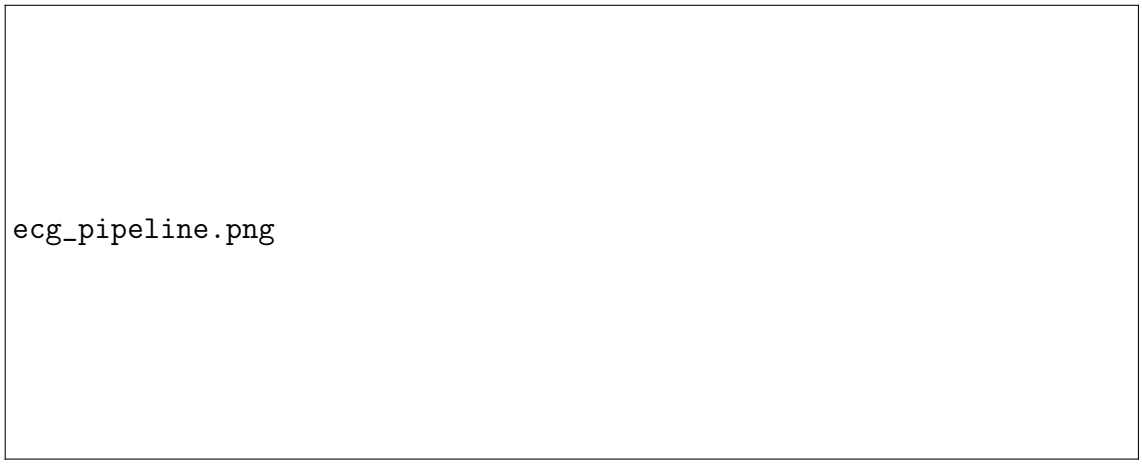


Figure 3.1: End-to-end system architecture linking the ESP32 edge node to the Android application.

3.2 Hardware Data Acquisition

An ESP32-S3 microcontroller is coupled with dual AD8232 analog front-end amplifiers providing high common-mode rejection ($>$80 dB). ECG biopotentials are sampled at 250 Hz across Lead I (pin A0) and Lead II (pin A1). Figure 3.2 shows the in-vivo experimental setup.

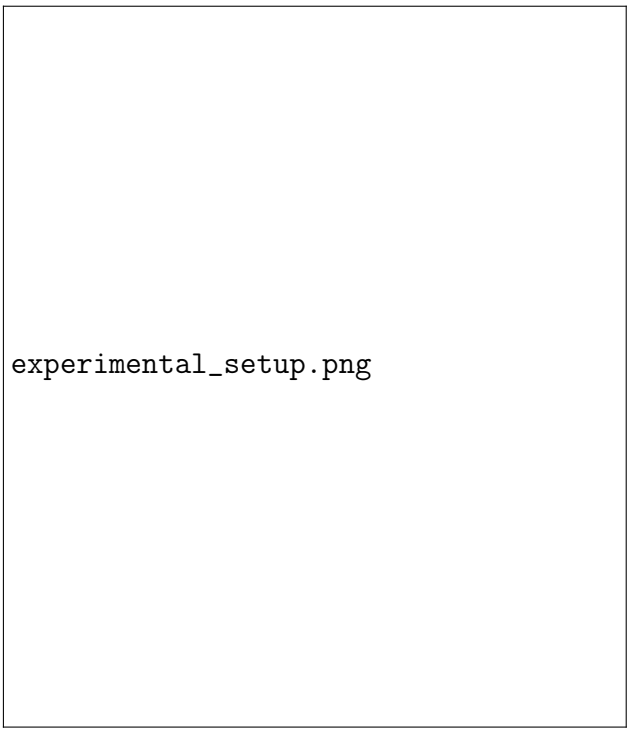


Figure 3.2: In-vivo experimental setup of the wearable ECG system during ambulatory testing.

3.3 Signal Quality Assessment (SQA)

Before neural inference, a real-time SQA algorithm evaluates each 5-second buffer ($N = 1250$ samples):

- **Flatline Detection:** Flags lead detachment when signal variance falls below an empirical floor ($\sigma^2 < \varepsilon_{flat}$).
- **Saturation Detection:** Identifies motion-induced ADC clipping:

$$C_{sat} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(|x[i]| \geq V_{rail}\right) \tag{3.1}$$

If severe distortion is detected, a **15-second inference lockout** is triggered—the AI engine halts and the user is alerted to adjust electrodes.

3.4 Preprocessing Pipeline

Segments passing the SQA gate undergo zero-phase 4th-order Butterworth high-pass filtering ($f_c = 1.0\,\text{Hz}$) followed by maximum absolute normalization to $[-1, 1]$:

$$x_{norm}[n] = \frac{x_{filt}[n]}{\max\left(|x_{filt}|\right)} \tag{3.2}$$

Figure 3.3 demonstrates the baseline stabilization.

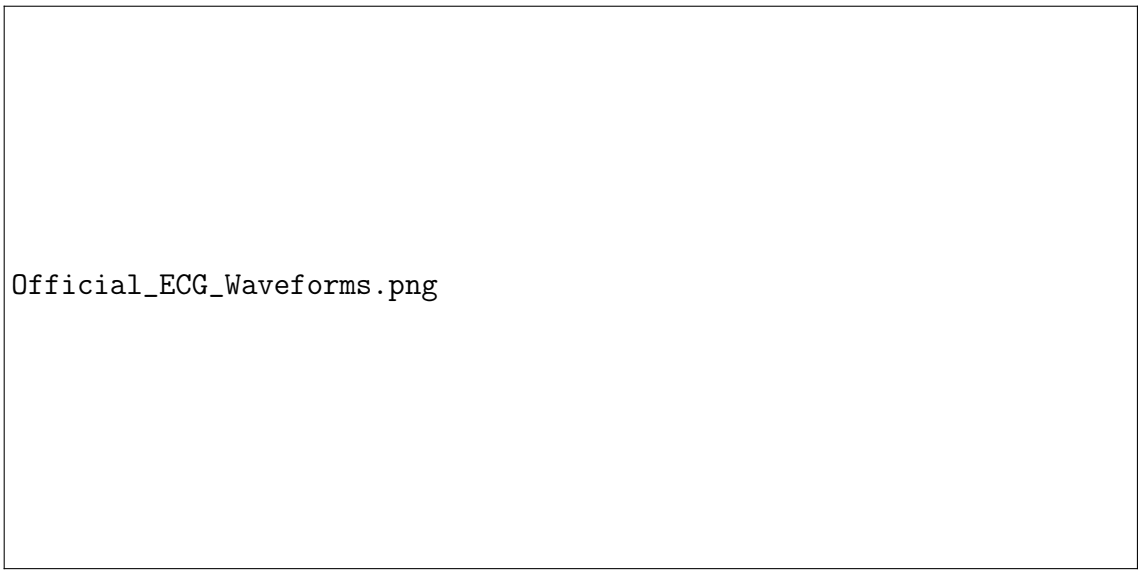


Figure 3.3: Raw ECG signals (top) versus filtered signals (bottom) demonstrating baseline stabilization.

3.5 Dataset and Validation Strategy

Training and evaluation data were sourced from PhysioNet databases (MIT-BIH Malignant Ventricular Arrhythmia Database, Creighton University VT Database). Records were strictly segregated by patient ID using **GroupShuffleSplit**, ensuring no patient in the test set appeared in the training set.

4. Implementation

4.1 Dual-Core Firmware Architecture

The firmware operates on the ESP32-S3 at 240 MHz under FreeRTOS, partitioning tasks across two CPU cores:

- **Core 1 (Real-Time Engine):** Runs the main `loop()` at 250 Hz. Every 4 ms, it reads ADC values from both leads, batches 5 samples into BLE packets (50 Hz notification rate), and executes real-time Pan-Tompkins R-peak detection for heart rate calculation.
- **Core 0 (Inference Engine):** Hosts the TFLite Micro thread. When Core 1 fills a 1250-sample buffer (5 seconds), an RTOS notification wakes Core 0 to perform SQA evaluation, Butterworth filtering, and CNN forward inference—all without blocking sampling.

4.2 1D-CNN Model Architecture

The neural network comprises 5 convolutional stages with a 16-unit embedding bottleneck and 3-class Softmax output (VT/VF, Normal, Noise). Total parameters: **43,219**.

Table 4.1: 1D-CNN Layer Configuration

Layer	Filters	Kernel	Activation	Pooling
Input	—	—	—	—
Conv1D-1	32	5	LeakyReLU	MaxPool(2)
Conv1D-2	64	5	LeakyReLU	MaxPool(2)
Conv1D-3	128	3	LeakyReLU	MaxPool(2)
Conv1D-4	128	3	LeakyReLU	MaxPool(2)
Conv1D-5	64	3	LeakyReLU	GlobalAvgPool
Dense	16	—	ReLU	—
Output	3	—	Softmax	—

The compiled FP32 C-byte array (`model.h`) occupies **182 KB** of flash memory. INT8 quantization was intentionally avoided because it degraded clinical sensitivity at morphological decision boundaries.

4.3 Native Android Application

A companion application was developed from scratch in Kotlin using Jetpack Compose:

- **BLE 5.0 Telemetry:** Maintains a persistent Bluetooth connection, parsing byte streams from two BLE characteristics—waveform data (Char 1) and AI predictions with vitals (Char 2).
- **Real-Time Canvas Rendering:** Hardware-accelerated continuous drawing of scrolling ECG waveforms with configurable gain and sweep speed.
- **HRV Analytics:** A rolling 20-beat FIFO buffer computes SDNN and RMSSD using adaptive MAD spike rejection.
- **Alert System:** Immediate visual and auditory alarms upon receiving a VT/VF classification from the edge.

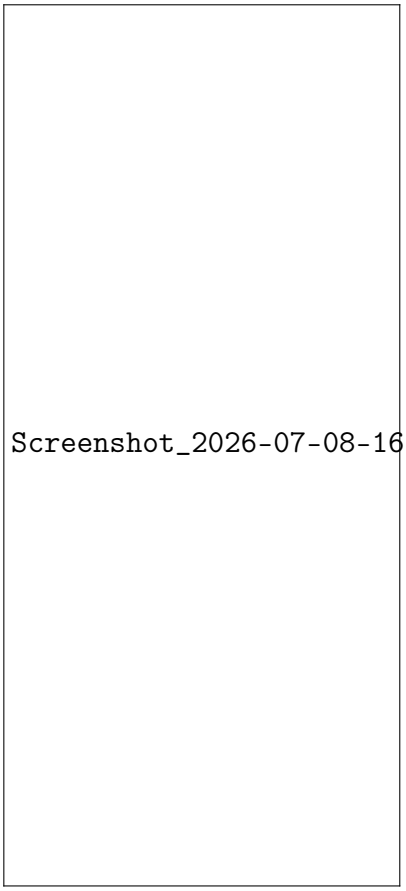


Figure 4.1: Android application showing real-time ECG waveform, arrhythmia alert status, and HRV metrics.

5. Results and Analysis

5.1 Classification Performance

The FP32 1D-CNN was evaluated strictly on unseen inter-patient test sets. The system achieved:

Table 5.1: Overall Diagnostic Performance

Metric	Value
Overall Accuracy	98.54%
VT/VF Sensitivity	97.46%
Specificity	99.72%
Inference Latency	< 2 ms
Model Size	182 KB
Parameters	43,219



Figure 5.1: Confusion matrix on unseen subjects across VT/VF, Normal, and Noise classes.

The confusion matrix (Figure 5.1) confirms near-perfect discrimination between genuine ventricular arrhythmias and physical noise artifacts.

5.2 Feature Space Visualization

The 16-dimensional embedding layer was visualized using t-SNE (Figure 5.2). The representations form tightly clustered, well-separated manifolds for all three classes, con-

firming strong generalization to unseen patients.

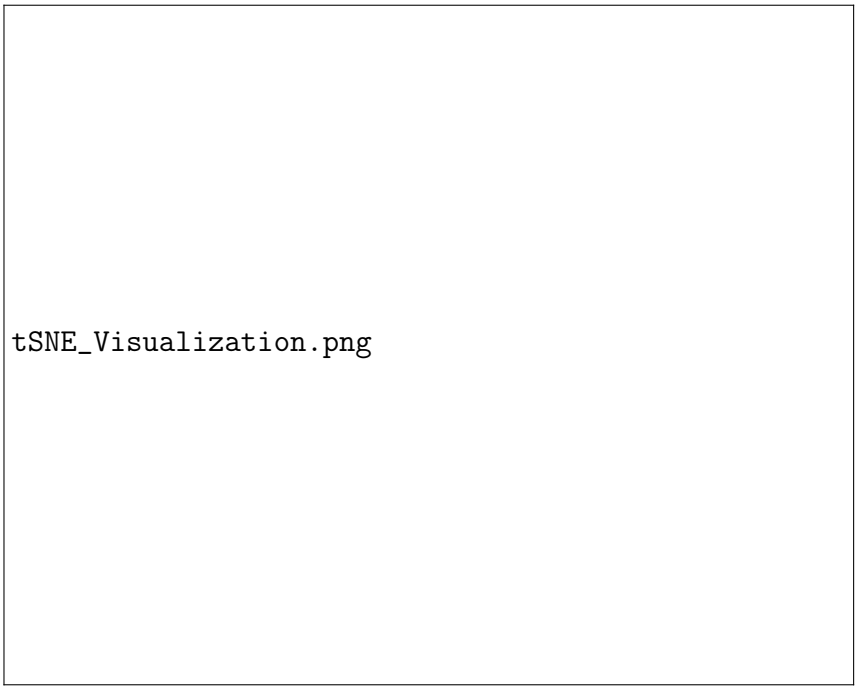


Figure 5.2: t-SNE visualization of 16D bottleneck embeddings showing distinct class separation.

5.3 Comparison with State-of-the-Art

Table 5.2 benchmarks the proposed system against recent VT/VF detection frameworks.

Table 5.2: Performance Comparison with State-of-the-Art Methods

Study	**Method**	**Edge**	**Acc (%)**	**Sens (%)**	**Spec (%)**
Picon *et al.* [6]	CNN-LSTM	×	93.10	94.20	91.80
Barbosa *et al.* [7]	Deep CNN	×	96.40	95.80	97.10
Han *et al.* [8]	CWT-STFT	✓	97.60	96.90	98.10
Mjahad *et al.* [9]	CNN-Fuzzy	×	97.20	96.50	97.90
Phukan *et al.* [10]	SQA-TinyML	✓	98.12	97.05	98.80
Proposed	**FP32 1D-CNN**	✓	**98.54**	**97.46**	**99.72**

The proposed system achieves the highest specificity (99.72%) and accuracy (98.54%) among all benchmarked works while maintaining a 182 KB footprint and sub-2 ms edge inference.

5.4 Individual Contributions

As the primary development intern, my contributions include:

1. Engineered the dual-core FreeRTOS C++ firmware establishing deterministic 250 Hz

sampling and background TFLite Micro inference on the ESP32-S3.

2. Designed and deployed the embedded SQA flatline/saturation detection and zero-phase Butterworth preprocessing pipeline.

3. Trained the lightweight 1D-CNN architecture in Python with subject-wise GroupShuffleSplit validation and exported the optimized FP32 C-array weights.

4. Developed the complete native Android application in Kotlin from scratch, integrating Jetpack Compose UI, BLE 5.0 parsing, and real-time HRV analytics.

6. Conclusions and Future Scope

6.1 Conclusions

This industrial training project successfully designed, implemented, and validated a cloud-independent edge-intelligent wearable system for real-time ventricular arrhythmia detection. The combination of hardware analog acquisition, dynamic SQA lockout, and an unquantized FP32 1D-CNN achieved **98.54%** accuracy and **99.72%** specificity on strictly unseen patients. Paired with a native Android application for real-time visualization and HRV analytics, the system delivers clinical-grade reliability in a wearable form factor.

6.2 Project Outcomes

- **Research Publication:** A formal research paper is being prepared for submission to a peer-reviewed IEEE journal.
- **PRAVAH 2026 Presentation:** The working prototype was demonstrated at the **PRAVAH 2026 Science & Innovation Fair** (24–25 June 2026) at IIT Kharagpur under the KRITI Initiative, as shown in Figure 6.1.



Figure 6.1: Demonstrating the prototype at PRAVAH 2026, IIT Kharagpur.

6.3 Future Scope

1. **Multi-Class Arrhythmia Detection:** Extend classification to Atrial Fibrillation (AFib), Atrial Flutter, and bundle branch blocks.

2. **Multi-Modal Sensor Fusion:** Integrate PPG sensors for continuous SpO2 and blood pressure estimation alongside ECG.

3. **Hardware Miniaturization:** Transition from breadboard modules to a flexible PCB bio-patch form factor for enhanced wearability.

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